Q4 – Sets: Elective Registration Clash Detection

Set operations

Use intersection (&) for students in both electives, symmetric difference (^) for students in exactly one, union (|) for everyone registered, and difference (-) from all_students for students in neither. Sets automatically remove duplicates and provide average O(1) membership tests.

Limitations

Sets are unordered and discard duplicate counts and original ordering. Convert results to sorted lists for deterministic reports. Lists cannot be set members because they are mutable; tuples of hashable values can be members.

Complexity

Nested loops cost O(n·m). Sorting both lists and merging costs O(n log n + m log m). Set construction and operations are O(n+m) average, with storage proportional to the number of unique students.

Python Program

```
ai_elective = {"192311001", "192311004", "192311007", "192311009"}
iot_elective = {"192311002", "192311004", "192311009", "192311010"}
all_students = {f"1923110{i:02d}" for i in range(1, 11)}
both = ai_elective & iot_elective
exactly_one = ai_elective ^ iot_elective
neither = all_students - (ai_elective | iot_elective)
print("clash      :", sorted(both))
print("exactly one:", sorted(exactly_one))
print("neither    :", sorted(neither))
```